

INFORMS 2026 Data Mining Society Data Challenge

Variable Descriptions: DM_Train.csv and DM_Test.csv

This document describes every variable in the training file (DM_Train.csv, 51,624 rows $\times$ 63 columns) and the test file (DM_Test.csv, 13,608 rows $\times$ 43 columns). Statistics are computed from the training data. The OSI formula and component derivations are in [osi_methodology.pdf](#). Weather variable sources and aggregation methods are in [weather_variables.pdf](#).

All variables listed in this document are present in both the train and test files, with two exceptions: (1) `split` and `severity_tier` are present in the train file only; (2) outage variables (`outageCount`, `outage_pct`, P_t , N_t , D_t , R_t , `osi`, and all lag columns) are present in both files but are **NaN for March 14–19 in the test file**. Weather variables and identifiers are present for all 216 hours in both files. OSI forecast targets are present in the train file only and withheld entirely from the test file.

Identifiers and Metadata

Do not use as model input features.

Column	Description
<code>timestamp_et</code>	Hourly timestamp in Eastern Time (EST, UTC−5). Ranges from 2026-03-11 00:00 to 2026-03-19 23:00; 216 unique timestamps, one row per county per hour.
<code>fipsCode</code>	FIPS county code. Uniquely identifies each county. Train: 239 counties. Test: 63 counties. No overlap between files.
<code>countyName</code>	Human-readable county name. 187 unique names across 302 counties (some repeat across states).
<code>stateName</code>	Full state name: Indiana, Ohio, Pennsylvania, West Virginia.
<code>stateAbbr</code>	Two-letter abbreviation: IN, OH, PA, WV.
<code>in_event_window</code>	False = March 11–12 pre-event lag period. True = March 13–19 storm window.
<code>split</code>	Always “TRAIN” in the training file. <i>Train file only.</i>
<code>severity_tier</code>	County severity level (0–4) within state, derived from peak OSI rank and average outage duration. Use for stratified cross-validation only; not a model input feature. <i>Train file only.</i>

Raw Outage Variables

Direct hourly measurements from PowerOutage.com. **NaN for March 14–19 in the test file.**

Column	Description
outageCount	Absolute customers without power this hour. Range: 0–97,457. Mean: 379. Zero for 43% of rows.
customersTracked	Total customers monitored in the county; denominator for all fractions. Range: 1,232–696,427. May vary slightly within a county due to multi-utility reporting timing; all fractions use the row-level value and are internally consistent. <i>Present for all 216 hours in both files.</i>
outage_pct	Current outage fraction as percentage (0–100). Computed as $\text{outageCount}/\text{customersTracked} \times 100$, capped at 100. Mean: 0.91%. Median: 0.01%.

OSI State and Flow Components

Four hourly variables used to construct the Outage Severity Index. See [osi_methodology.pdf](#) for full formula derivation. **NaN for March 14–19 in the test file.**

Column	Description
P_t	Current outage fraction. $P_t = \text{outageCount}/\text{customersTracked}$. Range: 0–1. Mean: 0.0091.
N_t	New outage growth rate this hour. $N_t = \max(0, \Delta\text{outageCount})/\text{customersTracked}$. Only positive when worsening. Range: 0–0.379. Zero for 59% of rows.
D_t	6-hour rolling mean of P_t . Captures sustained persistence vs recent spike. Range: 0–1. Mean: 0.0092.
R_t	Restoration rate this hour. $R_t = \max(0, -\Delta\text{outageCount})/\text{customersTracked}$. Only positive when improving. Range: 0–0.204. Zero for 52% of rows. Subtracts from OSI; actively recovering counties score lower.
osi	Hourly OSI composite score. Range: 0–0.65. Mean: 0.0062. Zero for 44% of rows. OSI cannot be reconstructed for March 14–19 test rows since the OSI components are withheld for that window; this is by design.

Event-Level Constant

Column	Description
event_duration_h	Observed storm window in hours. Always exactly 168 for every county (March 13–19 = 7 days). Constant across all rows. Carries no predictive information.

Lag Features

Pre-computed historical features. NaN appears at the start of each county's sequence where the lookback extends before the dataset begins (1, 3, 6, 24, or 48 NaN rows per county respectively). **NaN for all March 14–19 rows in the test file.**

Column	Description
outage_pct_lag1h	Outage % 1 hour ago. NaN for first 1 row per county. Strongest short-horizon predictor of current outage state.
outage_pct_lag3h	Outage % 3 hours ago. NaN for first 3 rows per county.
outage_pct_lag6h	Outage % 6 hours ago. NaN for first 6 rows per county.
outage_pct_lag24h	Outage % 24 hours ago (same time yesterday). NaN for first 24 rows per county. Particularly informative for the two-storm structure; reveals county state during first storm wave.
outage_pct_lag48h	Outage % 48 hours ago. NaN for first 48 rows per county. For storm-onset hours, draws on the pre-event window and reveals baseline state.
osi_lag1h	OSI 1 hour ago. NaN for first 1 row per county. Richer than outage_pct_lag1h because it encodes flow history.
osi_lag3h	OSI 3 hours ago. NaN for first 3 rows per county.
osi_lag6h	OSI 6 hours ago. NaN for first 6 rows per county.
osi_lag24h	OSI 24 hours ago. NaN for first 24 rows per county. Key for detecting two-storm interaction; high osi_lag24h means county already hit by first wave.
osi_lag48h	OSI 48 hours ago. NaN for first 48 rows per county.

Prediction Targets

Withheld entirely from the test file. In the training file, NaN appears for the last k rows per county per horizon (last 1, 6, 24, and 48 rows respectively) where the future window extends beyond the dataset end.

Column	Description
osi_target_t01h	OSI 1 hour ahead. Range: 0–0.599. Mean: 0.0062. NaN for last 1 row per county (239 NaN total).
osi_target_t06h	OSI 6 hours ahead. Range: 0–0.599. Mean: 0.0063. NaN for last 6 rows per county (1,434 NaN total). Operationally: crew pre-positioning window.
osi_target_t24h	OSI 24 hours ahead. NaN for last 24 rows per county (5,736 NaN total). Operationally: next-day mutual aid planning.
osi_target_t48h	OSI 48 hours ahead. Range: 0–0.598. Mean: 0.0071. NaN for last 48 rows per county (11,472 NaN total). Most independent horizon (correlation with t+1h: 0.056).

OSI Delta Columns: Optional Alternative Targets (train file only)

$\text{osi_delta_tXXh} = \text{osi_target_tXXh} - \text{osi}$ (change from current to future OSI). Not scored directly but equivalent to the targets after adding back current OSI. Provided as a convenience for participants who prefer to model change rather than level. Same NaN pattern as the corresponding target.

Column	Description
osi_delta_t01h	Change in OSI to t+1h. Range: -0.369 – 0.596 .
osi_delta_t03h	Change in OSI to t+3h. Not a scored horizon but retained for reference.
osi_delta_t06h	Change in OSI to t+6h. Range: -0.369 – 0.596 .
osi_delta_t24h	Change in OSI to t+24h.
osi_delta_t48h	Change in OSI to t+48h. Range: -0.640 – 0.593 . Wider range reflects greater 48-hour uncertainty.

Weather Features (22 columns, all 216 hours in both files)

All 22 weather variables are present for all 216 hours in both the train and test files. Full descriptions of sources, units, aggregation methods, and data quality notes are in `weather_variables.pdf`.

Important: `wind_dir_10m` is circular (0 – 360°). Do not treat as linear. Encode as: $\text{wind_sin} = \sin(2\pi \times \text{wind_dir}/360)$, $\text{wind_cos} = \cos(2\pi \times \text{wind_dir}/360)$.

Column	Source		Unit	Description
gust	URMA km	2.5	mph	Wind gust at 10 m. Strongest single predictor of outage onset.
wind_speed_10m	URMA km	2.5	mph	Sustained wind speed at 10 m.
wind_dir_10m	URMA km	2.5	degrees	Wind direction at 10 m. Circular; encode as sin/cos.
t2m	URMA km	2.5	Kelvin	Air temperature at 2 m. Determines rain vs snow/ice.
d2m	URMA km	2.5	Kelvin	Dewpoint at 2 m. Used to derive humidity and precipitation type.
sp	ERA5 km	31	hPa	Surface pressure.
mslma	ERA5 km	31	hPa	Mean sea-level pressure.
blh	ERA5 km	31	m	Boundary layer height. High values indicate convective activity.
tp	ERA5 km	31	mm	Total precipitation (rain + snow).
rain	ERA5 km	31	mm	Liquid rainfall only.
csnow	ERA5 km	31	cm	Snowfall. Compounds outage severity via line loading.
sdwe	ERA5 km	31	m	Snow depth (water equivalent). Pre-existing snowpack.
tcc	ERA5 km	31	%	Total cloud cover.
lcc	ERA5 km	31	%	Low cloud cover (below ~2 km).
mcc	ERA5 km	31	%	Mid cloud cover (2–6 km).
hcc	ERA5 km	31	%	High cloud cover (above 6 km). May precede fronts by 12–24 h.
sdsurf	ERA5 km	31	W/m ²	Surface downwelling shortwave radiation. Zero at night.
direct_rad	ERA5 km	31	W/m ²	Direct solar radiation component.
diffuse_rad	ERA5 km	31	W/m ²	Diffuse radiation. Remains non-zero under cloud cover.
r2	ERA5 km	31	%	Relative humidity at 2 m. High + near-freezing = icing risk.
vpd	ERA5 km	31	kPa	Vapour pressure deficit.
et0	ERA5 km	31	mm/hr	Reference evapotranspiration.
soil_moist	ERA5 km	31	m ³ /m ³	Near-surface soil moisture. High values reduce tree root-holding strength, increasing wind-throw risk.